Organic Food Supply-Chain Optimization.

1 Problem

A produce company supplies organically grown apples to four regional specialty stores. After the apples are collected at the company's orchard, they are transported to any of three preparation centers where they are prepared for retail (by undergoing extensive cleaning and then packaging). After the apples are prepared, they are shipped to the specialty stores. Due to the fragility of the product, the cost of transporting this organic fruit is rather high. The company has three preparation centers available for use. The tables below contains the following: (i) the unit transportation costs (in $/pound) to get the apples from company's orchard to the preparation centers, (ii) the cost ($/pound) to prepare the apples in the preparation centers, (iii) the preparation centers' monthly capacities, (iv) the demand at the specialty stores, and (v) the unit costs of transporting the apples from the preparation centers to the specialty stores.

2 Data

Preparation Center	**Transportation Cost ($/pound) (Orchard to Preparation Center)**	**Preparation Cost ($/pound)**	**Monthly Capacity (pounds)**
1	$0.45	$0.15	300
2	$1.00	$0.20	500
3	$1.62	$0.18	800

Shipping Cost ($/pound)				
Preparation Center	**Organic Orchard**	**Fresh & Local**	**Healthy Pantry**	**Season's Harvest**
1	$0.80	$1.10	$0.70	$1.40
2	$1.20	$1.10	$0.50	$1.40
3	$0.20	$1.40	$1.30	$1.70
Monthly Demand (pounds)	300	500	400	200

Let

$\boldsymbol{P} = \{Preparation\ Center\ 1, Preparation\ Center\ 2, Preparation\ Center\ 3\}$

$\boldsymbol{S} = \{Organic\ Orchard, Fresh\ \&\ Local, Healthy\ Pantry, Season's\ Harvest\}$

$\boldsymbol{capacity_i} = Monthly\ capacity\ of\ each\ preparation\ center\ i\ (in\ pounds)\ , i \in P$

$\boldsymbol{demand_i} = Monthly\ demand\ in\ pounds\ in\ store\ j\ , j \in S$

$\boldsymbol{Transportation\ Cost} = The\ cost\ involved\ in\ transportation\ from\ the\ orchard\ to\ the$

$preparation\ center\ i, for\ i \in P$

$\boldsymbol{Preparation\ Cost} = \ The\ cost\ involved\ in\ preparing\ apples\ for\ the\ retail\ store, i \in P$

$\boldsymbol{Shipping\ Cost} = \ The\ cost\ of\ delivering\ apples\ from\ preparation\ center\ i\ to\ store,$
$i \in P\ \ j \in S$

$\boldsymbol{C_{ij}} \quad = \ Total\ cost\ of\ transportation, preparation, and\ shipping\ for\ apples$
$from\ preparation\ center\ i\ to\ store\ j\ , i \in P\ , j \in S.$

$$x_{11} \ = \ Amount\ of\ apples\ sent\ to\ Organic\ Orchard\ from\ Preparation\ Center\ 1$$

$$x_{12} = \ Amount\ of\ apples\ sent\ to\ Fresh\ \&\ Local\ from\ Preparation\ Center\ 1$$

$$x_{13} = \ Amount\ of\ apples\ sent\ to\ Healthy\ Pantry\ from\ Preparation\ Center\ 1$$

$$x_{14} \ = \ Amount\ of\ apples\ sent\ to\ Season's\ Harvest\ from\ Preparation\ Center\ 1$$

$$x_{21} \ = \ Amount\ of\ apples\ sent\ to\ Organic\ Orchard\ from\ Preparation\ Center\ 2$$

$$x_{22} = \ Amount\ of\ apples\ sent\ to\ Fresh\ \&\ Local\ from\ Preparation\ Center\ 2$$

$$x_{23} \ = \ Amount\ of\ apples\ sent\ to\ Healthy\ Pantry\ from\ Preparation\ Center\ 2$$

$$x_{24} = \ Amount\ of\ apples\ sent\ to\ Season's\ Harvest\ from\ Preparation\ Center\ 2$$

$$x_{31} \ = \ Amount\ of\ apples\ sent\ to\ Organic\ Orchard\ from\ Preparation\ Center\ 3$$

$$x_{32} = \ Amount\ of\ apples\ sent\ to\ Fresh\ \&\ Local\ from\ Preparation\ Center\ 3$$

$$x_{33} \ = \ Amount\ of\ apples\ sent\ to\ Healthy\ Pantry\ from\ Preparation\ Center\ 3$$

$$x_{33} = \ Amount\ of\ apples\ sent\ to\ Season's\ Harvest\ from\ Preparation\ Center\ 3$$

3 Objective in Words

Decide how many pounds of apples to transport from the orchard to each preparation center and then to each specialty store so that the total cost is minimized, subject to the following constraints:

- The amount of apples to be prepared in each preparation center must not exceed its monthly capacity.
  - Capacity of the apples to be prepared in the preparation center 1 ≤ 300.
  - Capacity of the apples to be prepared in the preparation center 2 ≤ 500.
  - Capacity of the apples to be prepared in the preparation center 3 ≤ 800.
- The total amount transported to each specialty store must meet its monthly demand.
  - The total demand of apples in store1 (organic orchard) ≥ 300
  - The total demand of apples in store2 (fresh and local) ≥ 500
  - The total demand of apples in store3 (healthy pantry) ≥ 400
  - The total demand of apples in store4 (seasons harvest) ≥ 200
- Non-negativity constraints exist for all Values.

4 Decision Variables

Let

$\boldsymbol{x_{ij}} = The\ amount\ of\ apples(in\ pounds)\ shipped\ from\ preparation\ center\ i\ to\ store\ j\ ,$
$for\ i\ \in\ P\ , j\ \in\ S$

$\boldsymbol{C_{ij}}\ = Total\ cost\ of\ transportation, preparation, and\ shipping\ for\ apples$
$from\ preparation\ center\ i\ to\ store\ j\ , i \in P\ , j \in S.$

5 Algebraic Formulation

$$Minimize\ \sum_{i \in P}. \sum_{j \in S} C_{ij} X_{ij} \qquad \text{(Total\_cost)}$$

$$Where, C_{ij} = Transportation\ Cost_i + Preparation\ Cost_i + Shipping\ Cost_i$$

Subject to:

$$\sum_{j \in S} x_{ij} \leq capacity_i\ , i \in P \qquad \text{(Capacity\_Constraint)}$$

$$\sum_{i \in P} x_{ij} \geq demand_j\ , j \in S \qquad \text{(Demand\_Constraint)}$$

$$x_{ij} \geq 0\ , i \in P\ , j \in S \qquad \text{(Non-Negativity Constraint)}$$

6 Implementation

An implementation and solution of the model using Python, AMPL is available here:

https://colab.research.google.com/drive/103DysO3FyWHUIIOxDRfytYgxQh5Aj0h3?usp=sharing

7 Results

Optimal Solution (Amount of Apples Shipped from Each Preparation Center to Each Store) are as follows:

Preparation Center	Fresh & Local	Healthy Pantry	Organic Orchard	Season's Harvest
1	100	0	0	200
2	100	400	0	0
3	300	0	300	0

This allocation results in Total Cost of: **$3,040.00**

8 ChatGPT Prompt:

Coaching a generative AI (I have used ChatGPT) to produce AMPL code to solve the problem is shown here:

https://chatgpt.com/share/66fdfdf0-81c8-8010-b704-f6e4e9307d14